

Instructions: Marks are shown in [x]. Justify your steps.
Results from class-work can be used, but quote precisely.
Result from Home-Assignments, if used, must be proved,
even if it is explained in class and hints given – because
I do not know whether you worked it out or not.

1. (a). If $0 < x < 1$, show it cannot have more than two different decimal expansions. [3]
(b). Expansion of a number to the base 8 is $0.212121212121 \dots$. Calculate its continued fraction expansion. [3]

2. (a). Find the radius of convergence of $\sum \frac{1}{(17)^n + (29)^n} x^n$ [3]
(b). (x_n) is a sequence of strictly positive numbers; $x_n \rightarrow 43$. Let $f(n) = n^2 - n + 1$ and $y_n = x_{f(n)}$. Does the sequence x_n/y_n converge? If so find its limit. [3]

3. Test for convergence

(i) $\sum \frac{n^e}{e^n}$ (ii) $\sum \frac{5000}{(1/2)^n + (3/4)^n + (9/10)^n}$ [6]

4. Show that the following series is convergent.

$$1 + \frac{1}{2} - \frac{1}{3} - \frac{1}{4} + \frac{1}{5} + \frac{1}{6} - \frac{1}{7} - \frac{1}{8} + \frac{1}{9} + \frac{1}{10} - \frac{1}{11} - \frac{1}{12} + \dots$$

two positive and two negative terms alternate. [6]

5. Consider the function $f(x) = 1 + x^2 + x^3$ on $[-1, 1]$. Show explicitly a number $\delta > 0$ such that $|f(x) - f(y)| < 0.1$ whenever $|x - y| < \delta$. (you should show your δ works). [6]

GOOD LUCK